Icepak Help

EC XML File Format

> **Note:** Only axis-aligned (orthogonal) geometries are exported.

Region

The following is the EC XML format for the region (computational domain).

```
<solutionDomain>
  <location>
    <x>0</x>
    <y>0</y>
    <z>0</z>
  </location>
  <size>
    <x>1</x>
    <y>1</y>
    <z>1</z>
  </size>
</solutionDomain>
```

Blower

Blower objects are not exported.

Plate

Note the following regarding exported plate objects:

- Adiabatic plates are exported with a thickness of 1 mm.
- Conducting plates with a **Thermal Specification** of **Conductance**, **Thermal Resistance**, or **Thermal Impedance** are exported with a specification of **Thickness** and given a 1 mm thickness.

The following is the EC XML format for a plate boundary condition and its geometry.

```
<geometry>
  <solid2dBlock>
    <name>ConductingPlate1_Rectangle1</name>
    <active>true</active>
    <location>
      <x>0</x>
      <y>-0.2</y>
      <z>0</z>
    </location>
    <size>
      <x>0.8</x>
      <y>0.8</y>
      <z>0.001</z>
    </size>
    <plane>+xy</plane>
    <material>Al-Extruded</material>
    <powerDissipation>0</powerDissipation>
  </solid2dBlock>
</geometry>
```

Wall

Note the following regarding exported wall objects:

- Symmetry walls are not exported.
- Walls are exported with a thickness of 1 mm.

The following is the EC XML format for a wall boundary condition and its geometry.

```
<geometry>
  <solid2dBlock>
    <name>StationaryWall1_Box1</name>
    <active>true</active>
    <location>
      <x>0.6</x>
      <y>0</y>
      <z>0</z>
    </location>
    <size>
      <x>0</x>
      <y>0.4</y>
      <z>0.4</z>
    </size>
    <plane>+yz</plane>
    <material>Al-Extruded</material>
    <powerDissipation>0</powerDissipation>
  </solid2dBlock>
</geometry>
```

Block (Box)

Note the following regarding exported block objects:

- All 3D shapes except for cylinders are exported as block objects.
- Hollow and fluid blocks are not exported.

The following is the EC XML format for a block boundary condition assigned to box geometry.

```
<geometry>
  <solid3dBlock>
    <name>Block1_Box1</name>
    <active>true</active>
    <location>
      <x>0.2</x>
      <y>0</y>
      <z>0</z>
    </location>
    <size>
      <x>0.2</x>
      <y>0.2</y>
      <z>0.2</z>
    </size>
    <material>Al-Extruded</material>
    <powerDissipation>0</powerDissipation>
  </solid3dBlock>
</geometry>
```

Block (Cylinder)

Hollow and fluid blocks are not exported.

The following is the EC XML format for a block boundary condition assigned to cylinder geometry.

```
<geometry>
  <solidCylinder>
    <name>Block1_Cylinder1</name>
    <active>true</active>
    <location>
      <x>0.117157287525</x>
      <y>-0.0828427124746</y>
      <z>0</z>
    </location>
    <size>
      <x>0.565685424949</x>
      <y>0.565685424949</y>
      <z>0.4</z>
    </size>
    <plane>+xy</plane>
    <material>Al-Extruded</material>
    <powerDissipation>0</powerDissipation>
  </solidCylinder>
</geometry>
```

Source (3D Box)

Note the following regarding exported 3D source objects:

- 3D sources are exported as source block objects.
- Sources with a **Thermal Condition** of **Surface Flux** or **Fixed Temperature** are exported with a power specification of 0.

The following is the EC XML format for a source boundary condition assigned to 3D Box geometry.

```
<geometry>
  <sourceBlock>
    <name>Source1_Box1</name>
    <active>true</active>
    <location>
      <x>0.6</x>
      <y>0.4</y>
      <z>0</z>
    </location>
    <size>
      <x>0.4</x>
      <y>0.1</y>
      <z>0.2</z>
    </size>
    <material>Al-Extruded</material>
    <powerDissipation>0</powerDissipation>
  </sourceBlock>
</geometry>
```

Source (2D Rectangle)

Note the following regarding exported 2D source objects:

- 2D sources are exported as 2D source block objects.
- Sources with a **Thermal Condition** of **Surface Flux** or **Fixed Temperature** are exported with a specification of power of 0.

The following is the EC XML format for a source boundary condition assigned to 2D rectangle geometry.

```
<geometry>
  <source2dBlock>
    <name>Source1_Rectangle1</name>
    <active>true</active>
    <location>
      <x>-0.2</x>
      <y>0.2</y>
      <z>0</z>
    </location>
    <size>
      <x>0.8</x>
      <y>0.2</y>
      <z>0.001</z>
    </size>
    <plane>+xy</plane>
    <material>Al-Extruded</material>
    <powerDissipation>0</powerDissipation>
  </source2dBlock>
</geometry>
```

Grille

Note the following regarding exported grille objects:

- The **Pressure Loss Type Loss Curve** is not supported. It is exported with a loss coefficient of 0 and a free area ratio of 1.
- The **Pressure Loss Type Loss Coefficient** is calculated based on the **Resistance Type**. For the given **Resistance Type**, the loss coefficient is computed as follow, where A is the free area ratio:
    - **Perforated Thin Vent**:

$$l_c = \frac{1}{A^2}\left[0.707(1-A)^{0.375} + 1 - A\right]^2$$

    - **Circular Metal Wire Screen**:

$$l_c = 1.3\,(1-A) + \left(\frac{1}{A} - 1\right)^2$$

    - **Two-Plane Screen Cyl. Bars**:

$$l_c = \frac{1.28\,(1-A)}{A^2}$$

The following is the EC XML format for a grille boundary condition and its geometry.

```
<geometry>
  <grille>
    <name>Grille1_Box1</name>
    <active>true</active>
    <location>
      <x>0.6</x>
      <y>0.2</y>
      <z>0</z>
    </location>
    <size>
      <x>0.001</x>
      <y>0.2</y>
      <z>0.2</z>
    </size>
    <plane>+yz</plane>
    <freeAreaRatio>1</freeAreaRatio>
    <lossCoefficient>0</lossCoefficient>
  </grille>
</geometry>
```

Resistance

Note the following regarding exported resistance objects:

- The **Pressure Loss Types Power Law** and **Loss Curve** are not exported.
- **Pressure Loss Types** of **Power Law** and **Loss Curve** are exported with a loss coefficient of 0 and a free area ratio of 1.
- For resistances with a **Pressure Loss Type** of **Device/Approach**, only the linear coefficient and free area ratio are exported.

The following is the EC XML format for a resistance boundary condition and its geometry.

```
<geometry>
  <flowResistance>
    <name>Resistance1_Box1</name>
    <active>true</active>
    <location>
      <x>0.2</x>
      <y>0.2</y>
      <z>0</z>
    </location>
    <size>
      <x>0.4</x>
      <y>0.2</y>
      <z>0.2</z>
    </size>
    <freeAreaRatio>
      <x>1</x>
      <y>1</y>
      <z>1</z>
    </freeAreaRatio>
    <lossCoefficient>
      <x>0</x>
      <y>0</y>
      <z>0</z>
    </lossCoefficient>
  </flowResistance>
</geometry>
```

Opening

Note the following regarding exported opening objects:

- Recirculating openings are not exported.
- Free openings are exported as grille objects with a loss coefficient of 0 and a free area ratio of 1.

Network

Note the following regarding exported network objects:

- Only one and two-resistor networks are exported.
- Network faces must be parallel and part of the same 3D geometry.

The following is the EC XML format for a network boundary condition and its geometry.

```
<geometry>
  <twoResistorModel>
    <name>Network1_Box1</name>
    <active>true</active>
    <location>
      <x>0.03</x>
      <y>0.02</y>
      <z>0.02</z>
    </location>
    <size>
      <x>40</x>
      <y>60</y>
      <z>20</z>
    </size>
    <material>Al-Extruded</material>
    <powerDissipation>5</powerDissipation>
    <thetaJC>5</thetaJC>
    <thetaJB>1</thetaJB>
  </twoResistorModel>
</geometry>
```

PCB

Note the following regarding exported PCB objects:

- Only PCB substrate layers and exported. Devices and solderballs are not exported.
- The default assigned material is FR-4.

The following is the EC XML format for a PCB component.

```
<geometry>
  <printedCircuitBoard>
    <name>pcb0</name>
    <active>true</active>
    <location>
      <x>-17.8308</x>
      <y>-16.51</y>
      <z>0</z>
    </location>
    <size>
      <x>35.6616</x>
      <y>33.02</y>
      <z>0.001</z>
    </size>
    <material>FR-4</material>
  </printedCircuitBoard>
</geometry>
```

Heat sink

Heat sink bases and pins are exported as individual solid 3D blocks with its bounding box size and location data.

The following is the EC XML format for a heat sink component.

> **Note:** The heat sink definition contains one solid 3D block entry for the base and each pin.

```
<geometry>
  <heatsink>
    <name>heatsink4</name>
    <active>true</active>
    <parts>
      <solid3dBlock>
        <name>base</name>
        <active>true</active>
        <location>
          <x>-0.1</x>
          <y>-0.1</y>
          <z>0</z>
        </location>
        <size>
          <x>0</x>
          <y>0</y>
          <z>0</z>
        </size>
        <material>Al-Extruded</material>
        <powerDissipation>0</powerDissipation>
      </solid3dBlock>
    </parts>
  </heatsink>
</geometry>
```

Fan (2D)

Note the following regarding exported 2D fan objects:

- Fans with **Flow Type** of **Fixed Volumetric** and **Curve** are exported.
- The **Flow Type Fixed Mass Flow** is not exported.
- Fans with a **Flow Type** of **Fixed Mass Flow**are exported with a fixed flow specification of 0.
- The **Flow Direction** is exported as *<plane></plane>* in the exported xml schema.

The following is the EC XML format for a 2D fan component.

```
<geometry>
  <rectangular2dFan>
    <name>fan0</name>
    <active>true</active>
    <location>
      <x>-0.15</x>
      <y>-0.15</y>
      <z>0</z>
    </location>
    <size>
      <x>0.3</x>
      <y>0.3</y>
      <z>0.001</z>
    </size>
    <plane>+xy</plane>
    <flowDefinition>
      <fanCurve>
        <fanCurvePoint>
          <pressure>0</pressure>
          <volumeFlow>30</volumeFlow>
        </fanCurvePoint>
        <fanCurvePoint>
          <pressure>0.01</pressure>
          <volumeFlow>0</volumeFlow>
        </fanCurvePoint>
      </fanCurve>
    </flowDefinition>
  </rectangular2dFan>
</geometry>
```

Fan (3D)

Note the following regarding exported 3D fan objects:

- Fans with **Flow Type** of **Fixed Volumetric** and **Curve** are exported.
- The **Flow Type Fixed Mass Flow** is not exported.
- Fans with a **Flow Type** of **Fixed Mass Flow**are exported with a fixed flow specification of 0.
- The **Flow Direction** is exported as *<plane></plane>* in the exported xml schema.
- For 3D fans with a hub size specified as 0, the exported value is calculated as 1/10th the size of the fan dimensions, where the fan dimension used is the diameter (circular fans) or the smaller of width or length (rectangular fans).

The following is the EC XML format for a 3D fan component with a **Curve** specified as the **Flow Type**.

```
<geometry>
  <axial3dFan>
    <name>fan0</name>
    <active>true</active>
    <location>
      <x>-0.15</x>
      <y>-0.15</y>
      <z>0</z>
    </location>
    <size>
      <x>0.3</x>
      <y>0.3</y>
      <z>0.05</z>
    </size>
    <plane>+yz</plane>
    <hubSize>0</hubSize>
    <flowDefinition>
      <fanCurve>
        <fanCurvePoint>
          <pressure>0</pressure>
          <volumeFlow>30</volumeFlow>
        </fanCurvePoint>
        <fanCurvePoint>
          <pressure>0.01</pressure>
          <volumeFlow>0</volumeFlow>
        </fanCurvePoint>
      </fanCurve>
    </flowDefinition>
  </axial3dFan>
</geometry>
```

The following is the EC XML format for a 3D fan component with a **Fixed Volumetric** specified as the **Flow Type**.

```
<geometry>
  <axial3dFan>
    <name>Fan1_1</name>
    <active>true</active>
    <location>
      <x>-0.15</x>
      <y>-0.15</y>
      <z>0</z>
    </location>
    <size>
      <x>0.3</x>
      <y>0.3</y>
      <z>0.05</z>
    </size>
    <plane>+xy</plane>
    <hubSize>0.02</hubSize>
    <flowDefinition>
      <fixedFlowRate>20</fixedFlowRate>
    </flowDefinition>
  </axial3dFan>
</geometry>
```

Mesh Region

Note the following regarding exported mesh regions:

- A mesh region is exported as an assembly.
- Mesh regions that are assigned to geometries are associated with those geometries when exported.

The following is the EC XML format for a mesh region.

```
<geometry>
  <assembly>
    <name>MeshRegion1</name>
    <active>true</active>
    <geometry>
      <solid3dBlock>
        <name>Box1</name>
        <active>true</active>
        <location>
          <x>1</x>
          <y>0.5</y>
          <z>0.5</z>
        </location>
        <size>
          <x>0.2</x>
          <y>0.3</y>
          <z>0.3</z>
        </size>
        <material>Al-Extruded</material>
        <powerDissipation>0</powerDissipation>
      </solid3dBlock>
    </geometry>
  </assembly>
</geometry>
```

Material

Note the following regarding exported materials:

- Only solid materials are exported.
- Thermal modifiers are not exported for **Mass Density** and **Specific Heat**.
- Thermal Conductivity is supported for **Simple** and **Anisotropic** types.
    - For **Simple** with a **Quadratic** thermal modifier where C2 = 0, it is exported as linear temperature-dependent where the coefficient is equal to $C1 * solidCond$ and TempRef is defined in kelvins.
    - For **Anisotropic**, the thermal conductivity is exported as orthotropic.

The following is the EC XML format for a material.

```
<materials>
  <material>
    <name>Al-Extruded</name>
      <density>2800</density>
      <specific_heat>900</specific_heat>
      <surfaceEmissivity>0</surfaceEmissivity>
      <thermalConductivity>
        <isotropic>
          <conductivity>205</conductivity>
        </isotropic>
      </thermalConductivity>
  </material>
  <material>
    <name>Au-Typical</name>
      <density>19281</density>
      <specific_heat>131</specific_heat>
      <surfaceEmissivity>0</surfaceEmissivity>
      <thermalConductivity>
        <linearTemperatureDependant>
          <conductivity>313</conductivity>
          <coeff>2504</coeff>
          <tref>297.15</tref>
        </linearTemperatureDependant>
       </thermalConductivity>
  </material>
</materials>
```

Monitor Point

The following is the EC XML format for a monitor point.

```
<monitorPoint>
  <name>pmon_mon_pt_1_1</name>
  <active>true</active>
  <location>
    <x>0.4</x>
    <y>0.5</y>
    <z>0.5</z>
  </location>
</monitorPoint>
```

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